Project Core Items

All three are up and working. Occasionally, lines published to Google Sheets will duplicate.

Explore Items

Automatic reaction is up and working. Weather via webhook is not working. I did not start this part until the final day, after waiting the requisite 2 hours for the API key (per OpenWeather docs) to come up, the test in Particle documentation still failed. Brief analysis shows that API 3.0 support has been deprecated, so I suspect the Particle sample web integration piece may have been deprecated (on the OpenWeather side). While it should work in limited scope, I opted not to dig into the issue since this was not a required piece and going in this level of blind seemed not the best use of time on the final day.

In order to add this functionality, I would create a new custom webhook. My initial instinct for this project is to also name it telemetry like my other webhook for Google Sheets. However, true application of this would depend on real world needs. For temperature sensing within a greenhouse, daily highs and lows would likely be enough for some logic within the system (likely a Particle Logic function) to determine if an additional flag needed to be raised. In a use case like this, I would have the event be its own named item since it would not need to be triggered every minute as the system currently has the telemetry event triggering. In the firmware code, I would create a function similar to the subscriptionHandler event from the Particle documentation on this subject, but I would change the loop function to have (for the example above) a daily check and run it once a day and store the high and low as variables stored locally and also push them along to the Google sheet and logs produced by the device for easier visibility.

Problems encountered

I struggle to say that there were many problems encountered. I found the documentation covered many cases within this document. One problem was listed above with the weather webhook. The other problem I encountered was that occasionally a line was duplicated to the Google Sheet. This primarily happened when the Google Sheet was erased when I would clean up the lines from 6k to 0 every day or so. Additionally, this would happen occasionally when re-flashing the device as well. I narrowed it down to these moments and have pegged it to connectivity errors and calls happening asynchronously and attempting to gracefully recover. Because data was not lost, I did not spend a large amount of time on this since in a real world instance missing data would be a larger issue than repeated data.

AI tooling

I primarily used AI as a planning tool. I did find an interesting moment where AI led me to the web IDE provided by Particle. My assumption was that I could use that code 1:1 to initialize my firmware code and rip it. I found that the types provided in that code were not compatible and needed updating for the example of flashing the LED on each device. I was looking at the “heartbeat” light not the D7 pin light and did not wait long enough for the communication to

begin/stop. I discovered this once I applied logic and looked at the documentation to see I was observing the wrong light and the type for setting which pin to use was incorrect.

Shortcuts, limitations, and recommendations

I used the documentation and examples provided on the Particle site quite a bit, which could be considered a shortcut, but I am not sure that is too much of one. Additionally, rather than sensing a change in temperature, the Boron just assumes an ambient temperature and during the demo I change the threshold to trigger the response by the Photon device. In the real world the threshold would be a relatively static value and the response would be triggered by a thermometer reading published by the Boron device.

At scale

Much of what I have produced would have to be rethought at scale. For the Boron devices we would need to reduce the publishing frequency to the MVP. This would be use case dependent, though because so much happens down stream from the publish it would need to remain relatively frequent. I would also want us to reduce the amount of data in each call to as little as possible, and possibly separate the temperature and humidity (and whatever other readings) publish events out since some will be more time critical than others to be published at the largest (most infrequent) rate possible to reduce calls.

This would be similarly reflected in the Photon. It would stop reporting every second and instead report at a lower interval PLUS when an alert triggers. I would also want to explore what sort of push notification could be triggered in the corresponding Logic function depending on the alert, but that was not an avenue I explored in the course of this project.

For the Ledgers, I would change from device to product specific scopes wherever possible, but with a larger scale, I would still have a few Ledgers made since temperature thresholds could change from location to location despite doing the same “basic” task.
For the Logic function, this would become significantly more robust. At a high level, this would do the major work to determine if the Photon needs to throw a new alert.

This project does not account for connectivity issues or latency issues. I would have to look to my team for the first time to see best practice for this. I would expect the two changes would occur in the firmware and in the Logic function. The Logic function seems the best place to handle the bulk of the work, but again, I would want to see best practice.

Finally, two aspects of storage are a problem at a larger scale. Google Sheets is good for testing and a small fleet. With larger production fleets, I would be hesitant to recommend it for a few reasons: it is not auditable, we would encounter rate limits (though I could adjust to account for this to some degree), and it is potentially less secure. I would recommend moving to a database or looking to the team for a recommendation and best practices. The other storage problem is the logs produced by each device and published to the Particle console since the Particle console only stores the data for the length of the session. AI seemed to suggest that this was being worked on but I did not see any definitive documentation of this in a roadmap, so I would want to validate this or add an additional integration for these alerts to be persisted.